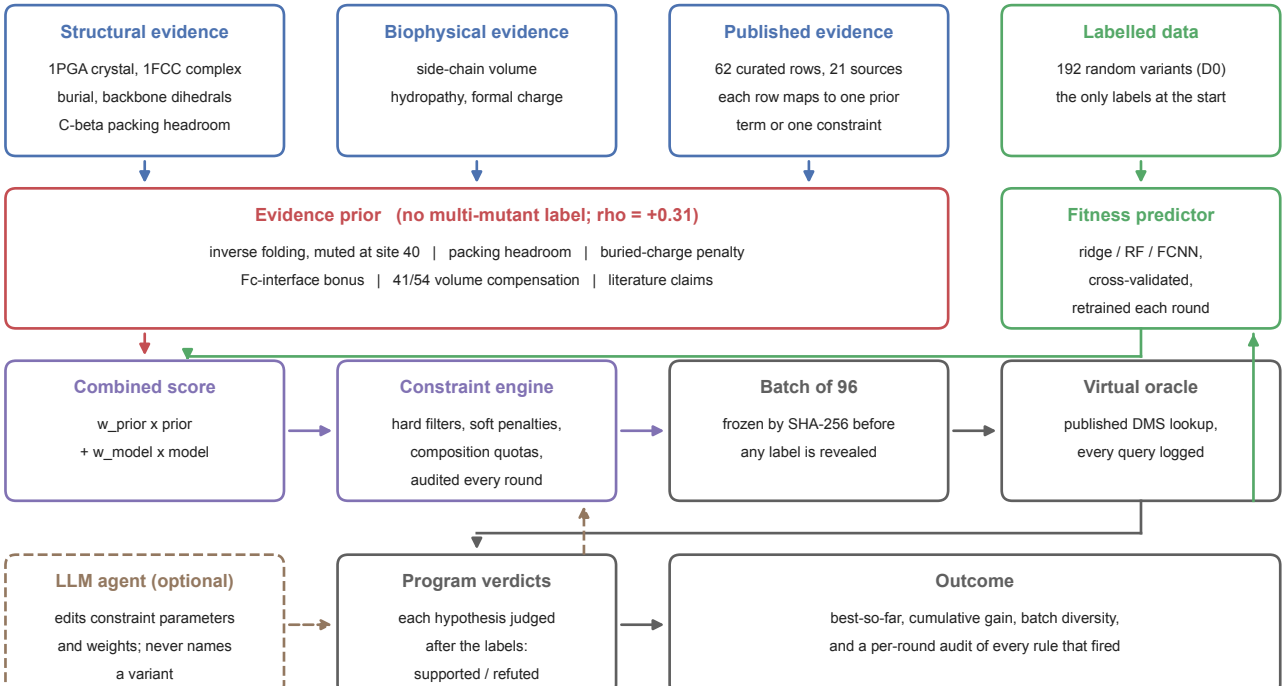
Structural evidence
1PGA crystal, 1FCC complex
burial, backbone dihedrals
C-beta packing headroom
Biophysical evidence
side-chain volume
hydropathy, formal charge
Published evidence
62 curated rows, 21 sources
each row maps to one prior
term or one constraint
Labelled data
192 random variants (D0)
the only labels at the start
Evidence prior (no multi-mutant label; rho = +0.31)
inverse folding, muted at site 40 | packing headroom | buried-charge penalty
Fc-interface bonus | 41/54 volume compensation | literature claims
Fitness predictor
ridge / RF / FCNN,
cross-validated,
retrained each round
Combined score
w_prior x prior
+ w_model x model
Constraint engine
hard filters, soft penalties,
composition quotas,
audited every round
Batch of 96
frozen by SHA-256 before
any label is revealed
Virtual oracle
published DMS lookup,
every query logged
LLM agent (optional)
edits constraint parameters
and weights; never names
a variant
Program verdicts
each hypothesis judged
after the labels:
supported / refuted
Outcome
best-so-far, cumulative gain, batch diversity,
and a per-round audit of every rule that fired